The Interquartile Range

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Introduction

The interquartile range (IQR) is the distance between the first and third quartiles, $Q_3 - Q_1$. It measures the width of the central 50% of the distribution and is robust to outliers, since extreme values do not affect the quartiles themselves. The IQR appears in boxplots, Tukey fences, and the median-IQR convention for reporting skewed biomedical variables.

Prerequisites

The reader should know what quantiles and quartiles are, and be comfortable with R's `quantile()` function.

Theory

For a sample, Q_1 and Q_3 are the 0.25 and 0.75 quantiles of the empirical CDF. The IQR is

$$\mathrm{IQR} = Q_3 - Q_1.$$

Tukey's hinges differ slightly from R's default quantiles; `fivenum()` returns the hinges and `IQR()` uses `quantile()` with `type = 7`.

For a normal distribution with SD σ, the IQR equals $2 \cdot 0.6745 \cdot \sigma \approx 1.349\sigma$. Equivalently, dividing the IQR by 1.349 gives a rough, robust estimate of the SD.

The **Tukey fences** at $Q_1 - 1.5 \cdot \mathrm{IQR}$ and $Q_3 + 1.5 \cdot \mathrm{IQR}$ define the whiskers of a standard boxplot; points beyond are flagged as outliers.

Assumptions

The IQR is purely descriptive and requires no distributional assumption.

R Implementation

```r
set.seed(2026)
x <- rnorm(80, mean = 75, sd = 12)

quantile(x, probs = c(0.25, 0.75))
IQR(x)

# Tukey fences
q1 <- quantile(x, 0.25); q3 <- quantile(x, 0.75)
fences <- c(lower = q1 - 1.5 * IQR(x),
            upper = q3 + 1.5 * IQR(x))
fences
sum(x < fences[1] | x > fences[2])

# Robust SD from IQR (approximately)
```

```r
IQR(x) / 1.349
sd(x)
```

Output & Results

```
  25%      75%
 66.4     83.1

[1] 16.7       # IQR

lower  upper
 41.4  108.1

[1] 12.4       # SD from IQR (robust)
[1] 11.9       # sample SD
```

The two SD estimates agree closely for approximately normal data; divergence signals heavy tails or outliers.

Interpretation

Report the IQR alongside the median for skewed data. A typical biomedical manuscript uses "median 6.8 days (IQR 4.2-10.1 days)" to summarise length-of-stay or similar variables.

Practical Tips

- The boxplot visually displays the IQR as the box; whiskers extend to the Tukey fences or to the nearest non-outlier data point.
- For very small samples ($n < 10$), the IQR is unstable; report the order statistics directly.
- The IQR is the scale estimator implicit in the Tukey boxplot; it is one of the most interpretable dispersion measures for audiences less familiar with SDs.
- In grouped summaries, compare IQRs across groups alongside medians; equal medians with different IQRs indicate different spreads at the same centre.
- Do not confuse IQR (dispersion) with inter-quartile ratio (Q_3/Q_1) occasionally used in log-skewed contexts.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Descriptive statistics and Table 1